UNICEF StartUp Lab | Safely Managed Sanitation Hackathon 2026

CONCEPT NOTE

FloodSafe WASH: Climate-Smart Sanitation Intelligence Platform

Team Name	FloodSafe WASH Team
Project Title	FloodSafe WASH: Climate-Smart Sanitation Intelligence Platform
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Submission Date	May 2026
Challenge	UNICEF StartUp Lab SMS Hackathon 2026

1. Executive Summary

FloodSafe WASH is a low-cost, offline-first digital platform that manages the full sanitation chain in Northern Ghana from containment through transport to treatment and reuse before, during, and after climate shocks such as flash floods and droughts.

The core innovation is a USSD and SMS-based coordination system that requires no smartphone, no internet connection, and no expensive hardware. Using existing mobile network infrastructure, toilet caretakers report fill levels via a simple dial code (*713#), weather forecast data triggers pre-emptive flood alerts 48 hours before heavy rains, and the nearest waste collector is automatically dispatched via SMS. This ensures waste is emptied and transported to treatment facilities before floodwater causes contamination.

Northern Ghana's White and Black Volta basin communities face compounding risks: high groundwater tables, fragile savannah soils, and zero coordination between communities, waste collectors, and treatment facilities. When floods hit, toilets overflow, raw waste enters water sources, children develop waterborne diseases, and girls drop out of school. FloodSafe WASH breaks this cycle by shifting from reactive emergency response to proactive, data-driven prevention.

The platform is designed to scale across the entire savannah belt, creating local sanitation jobs, supplying compost to farmers, and building a community-owned data infrastructure that improves with every flood season.

2. Technical Design & Climate Resilience

2.1 Flood & Drought Adaptation

FloodSafe WASH addresses both flood and drought scenarios through a dual-mode design:

Flood Mode (Primary Threat):

- Integrates Open-Meteo weather API (free, no API key required) to monitor 48-hour rainfall forecasts for Northern Ghana coordinates
- When forecast exceeds a threshold (>30mm/24hrs), the system triggers an automated SMS alert to all registered caretakers and collectors
- Alert instructs caretakers to request urgent collection before flood onset
- Fill levels reported via USSD are cross-referenced with flood risk to prioritise which toilets are emptied first
- System logs all pre-flood collections, creating evidence of climate response for funders and government

Drought Mode (Secondary Threat):

- During dry seasons, reduced rainfall increases manual waste transportation difficulties
- System shifts to weekly reporting cycles, conserving collector fuel and effort
- Fill-level data helps predict when pit latrines are approaching capacity, enabling planned desludging before crisis point

2.2 Infrastructure & Tech Stack

Layer	Technology	Reason
USSD Gateway	Africa's Talking API	Free sandbox; works on all Ghanaian networks (MTN, Vodafone, AirtelTigo)
SMS Alerts	Africa's Talking SMS	Bulk SMS at GHC 0.04/message; no smartphone needed
Weather Data	Open-Meteo API	Free, no API key, covers Ghana with hourly forecasts
Backend	Node.js / Python Flask	Lightweight; deployable on free-tier Render or Railway hosting
Database	SQLite / PostgreSQL	SQLite for prototype; PostgreSQL for scale
Frontend	Plain HTML/CSS/JS	Simple admin dashboard; no framework dependency
Hosting	Render.com (free tier)	Zero-cost deployment for prototype phase

3. Data Strategy & Operational Logic

3.1 Addressing Missing Data

Northern Ghana faces significant data gaps no existing groundwater maps, no soil permeability records, and no sanitation infrastructure database. FloodSafe WASH is designed to function without this data and to generate it over time.

How we operate without existing data:

- We use rainfall forecast data (freely available via Open-Meteo) as a proxy for flood risk, rather than requiring groundwater data
- Fill levels are reported by human caretakers, eliminating the need for expensive soil sensors
- Community-reported GPS coordinates are used to build a live toilet registry this is our baseline dataset, created by the platform itself
- Each USSD interaction logs: toilet ID, fill level, date/time, and collection outcome building a local dataset from Day 1

Data generation plan:

- Month 1: Register all pilot toilets with GPS coordinates and household count
- Month 2: Begin logging fill cycles how fast each toilet fills, correlated with rainfall data
- Month 3: Use fill-cycle data to generate predictive fill models for each toilet, calibrated to local groundwater behaviour
- Year 1: Share aggregated data with CWSA and District Assemblies to inform infrastructure planning

3.2 Maintenance & Post-Shock Recovery

System maintenance is designed to be community-led and low-cost:

Component	Responsible Party	Recovery Plan
USSD reporting	Toilet caretaker	Caretaker retraining takes 30 minutes; backup caretaker registered for each toilet
SMS dispatch	FloodSafe admin	Africa's Talking has 99.9% uptime SLA; fallback to manual phone calls
Weather alerts	Automated (Open-Meteo)	System auto-retries every 6 hours; no human intervention needed
Collector network	Local waste operators	Minimum 2 collectors registered per community; load-balanced dispatch
Server uptime	Render.com hosting	Free-tier hosting with automatic restarts; data backed up daily to Google Drive

4. Social Equity & Inclusion

4.1 Prioritising Vulnerable Groups

FloodSafe WASH was designed with Northern Ghana's most vulnerable groups at the centre of every decision:

Group	How FloodSafe WASH Serves Them
Children under 5	Pre-flood collection prevents waterborne disease that disproportionately kills young children

Girls in school	Safe school sanitation during floods reduces absenteeism and dropout rates
Women & caregivers	USSD works on basic phones used by women in rural areas; no literacy required for voice prompts
Persons with disabilities	SMS confirmations ensure collectors arrive on schedule, reducing uncertainty for mobility-impaired users
Elderly households	Automated dispatch removes the burden of physically seeking waste collection help

4.2 Affordability & Access

FloodSafe WASH is designed to be affordable at every level of the system:

For households:

- USSD reporting is free for users cost is borne by the platform (approx. GHC 0.01 per session)
- No smartphone, data plan, or app download required
- Works on GHC 5 basic phones common across Northern Ghana

For District Assemblies:

- Platform operating cost is approximately GHC 200/month for 50 toilets less than the cost of one emergency desludging trip
- Open-source codebase means no licensing fees
- Data generated can replace expensive manual sanitation surveys

For waste collectors:

- Coordinated dispatch reduces wasted trips and fuel costs
- Guaranteed collection requests create predictable income streams
- Platform creates formal records of service delivery for District Assembly payment

5. Implementation Roadmap & Risks

5.1 Implementation Phases

Phase	Key Activities	Deliverable
3-Day Bootcamp (Hackathon)	Build working USSD flow using Africa's Talking sandbox. Set up toilet registry with 5 sample entries. Connect weather API and trigger test flood alert. Demo full cycle: caretaker reports → alert fires → collector dispatched → SMS confirmation.	Working prototype with live USSD demo, weather integration, and SMS dispatch
Month 1 (Pilot Setup)	Select 1 pilot school in flood-prone community. Register 3-5 toilets in the system. Train	Registered pilot site with trained operator

	caretaker on USSD reporting. Partner with 1 local waste collector.	
Month 2 (Testing)	Run live USSD reporting cycles. Simulate flood alert scenario. Measure time from alert to collection. Identify friction points.	Test report with response time data
Month 3 (Measurement)	Track fill reports vs. confirmed collections. Survey caretakers and community. Document impact and lessons learned.	Impact report + scaling roadmap

5.2 Risk Mitigation

Risk	Likelihood	Mitigation Strategy
Low caretaker adoption of USSD	Medium	Conduct 30-minute hands-on training with practice sessions. Provide laminated USSD prompt cards. Register backup caretaker for each toilet.
Africa's Talking API downtime	Low	Fallback to direct phone call list maintained offline. System logs pending alerts and retries automatically.
Waste collector non-response	Medium	Register minimum 2 collectors per community. Non-response triggers escalation SMS to District Assembly sanitation officer.
Weather API inaccuracy	Low	Use 3-day rolling forecast average, not single-day predictions. Community-reported flooding can also trigger manual alerts.
Funding gap after pilot	Medium	Apply to UNICEF Innovation Fund and GSMA Mobile for Development. District Assembly can budget platform costs as preventive health expenditure.
Team member dropout	Low	All technical documentation maintained on GitHub. Modular codebase means any developer can continue work.

FloodSafe WASH — Securing the chain. Protecting the future.

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